FAST National University

Process Mining and Simulation
SE4009

Assignment 2

Cluster Analysis & Association Rule Learning

Instructor:
Dr. Isma ul Hassan

March 13, 2026

Submission Guidelines

Late submissions will NOT be accepted under any circumstances.

Submission Format

Submit a single `.zip` archive named:

`member1-id_member2-id.zip` (extend for 3–4 members)

The archive must contain:

- `report/` — PDF report with all analysis, figures, and interpretations.
- `code/` — All scripts or notebooks used to produce results.

Part 1: Dataset Selection

5 Marks

Select a **real-world dataset** of your choice. Record your dataset URL in the shared Google Sheet at the link below. Each group must register a unique dataset.

Open Dataset Registration Sheet

Do not use the sample dataset provided in the sheet.
The sample is for illustration only. Your group must select an independent dataset.

Dataset Requirements

- Publicly available with a valid, accessible URL.
- Contains multiple **numeric features** suitable for clustering.
- Has sufficient data points to produce meaningful clusters.
- Does not duplicate another group's registered dataset.
- Provide a short paragraph in your report describing the dataset and justifying your choice.

TOTAL: 5 Marks

Part 2: K-Means Clustering

20 Marks

Apply K-Means clustering to your dataset using the **Euclidean distance** metric.

Euclidean Distance Formula

$$d_{\text{euc}}(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

where $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ are two observations in n-dimensional feature space.

2.1 Distance Matrix

3 Marks

- Compute pairwise Euclidean distances between all observations (or a representative sample for large datasets).
- Present the distance matrix as a formatted table or a heatmap.

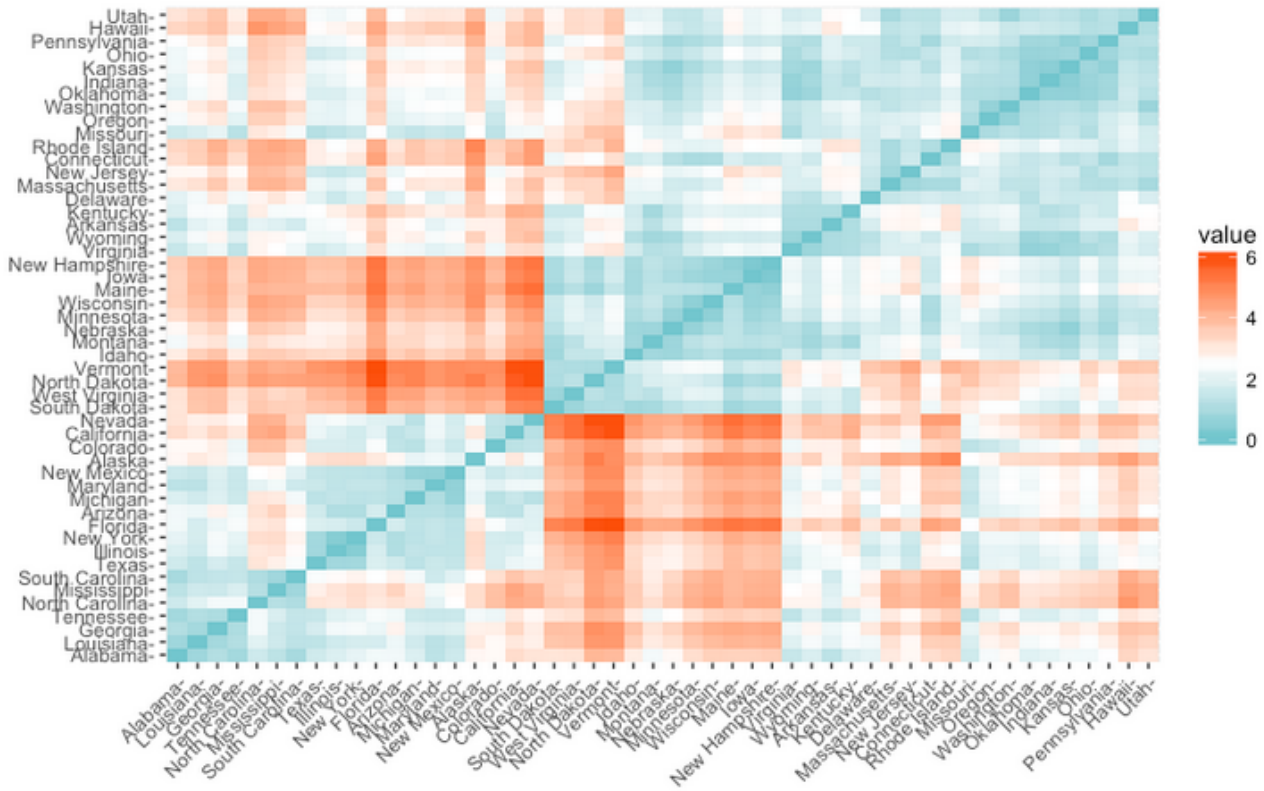

Figure 1: Distance Matrix

2.2 Centroid Initialisation

2 Marks

- Choose initial centroids either **randomly** or using **custom** / **domain-informed** positions.
- Clearly state and justify the initialisation strategy used.

2.3 Determining the Optimal Number of Clusters

5 Marks

- Run K-Means for at least **three different values of** k.
- Apply the **Elbow Method** or another justified criterion (e.g. Silhouette Score, Gap Statistic) to select optimal k.
- Show the diagnostic plot and state your chosen k with explanation.

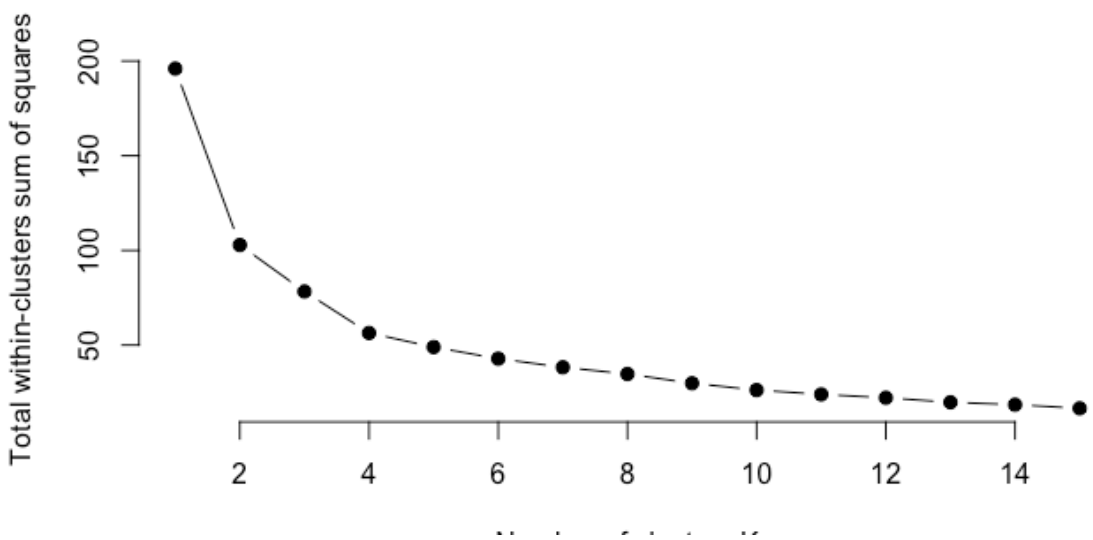

Figure 2: Elbow Method Visualization

2.4 Cluster Visualisation

5 Marks

- Produce **cluster maps** (scatter plots coloured by cluster) for each value of k tested.
- Where applicable, produce a domain-specific visualisation (e.g. a map with each region shaded by its cluster).

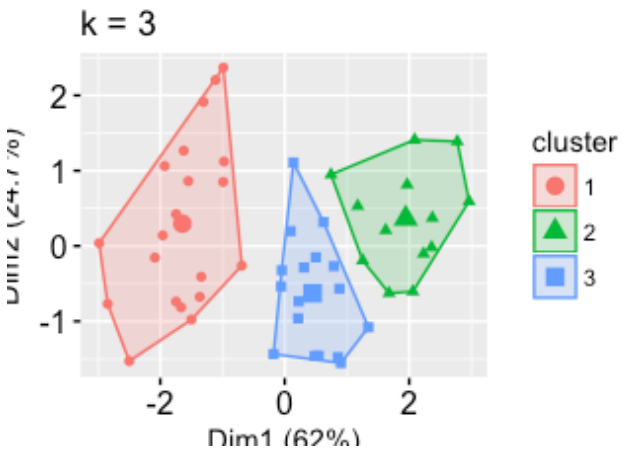

Figure 3: Cluster Map

2.5 CSV Output

3 Marks

- Export a `.csv` file appending a `Cluster` column to the original data, recording each point's cluster assignment.
- Include a snippet (first 10–15 rows) of this CSV in your report.

2.6 Per-Cluster Visualisation & Interpretation

2 Marks

- Visualise feature distributions **within each cluster** (e.g. box plots, violin plots, or bar charts of cluster means).
- Describe in writing what distinguishes each cluster from the others.

TOTAL: 20 Marks

Part 3: Hierarchical Clustering

20 Marks

Hierarchical algorithms build a nested sequence of clusters using previously established groupings. Two complementary strategies are required.

Agglomerative (Bottom-Up)

Starts with every observation as its own cluster, then **merges** the two closest clusters at each step until one cluster remains.
*Good at identifying **small, fine-grained** clusters.*

Divisive (Top-Down)

Starts with all observations in one cluster, then **splits** recursively into smaller clusters until each observation stands alone.
*Good at identifying **large, coarse** clusters.*

3.1 Agglomerative Clustering

8 Marks

Implementation

Apply agglomerative (bottom-up) hierarchical clustering. Specify and justify the linkage criterion used (single, complete, average, or Ward's method).

Dendrogram

Plot the complete dendrogram. Annotate the cut level used to obtain the final partition.

Cluster Map

Visualise the resulting clusters on a scatter plot or domain-specific map.

Interpretation

Describe the cluster structure revealed. Which small or tight clusters did the agglomerative approach identify? How do results compare to K-Means?

3.2 Divisive Clustering

8 Marks

Implementation

Apply divisive (top-down) hierarchical clustering. Describe the splitting criterion or algorithm used (e.g. DIANA or bisecting K-Means).

Split Diagram

Visualise the hierarchical splitting process (dendrogram or tree diagram). Annotate where the tree is cut to produce the final clusters.

Cluster Map

Visualise the resulting clusters.

Interpretation

Describe the large-cluster structure revealed. How does the divisive approach differ from the agglomerative results?

3.3 Comparison: Agglomerative vs. Divisive

4 Marks

- Present a **side-by-side** visualisation of both cluster assignments.
- Summarise key differences in cluster membership, compactness, and size.
- Conclude which method better suits your dataset and justify your answer.

TOTAL: 20 Marks

Qestion No 02: Association Rule Learning(Part 4)

30 Marks

Handwritten Requirement: This entire Part must be completed **individually by hand** by each group member as assigned below. Scan or photograph your handwritten work and include it in the submission ZIP under `report/handwritten/`.

Scenario: Software Quality & Metrics Analysis

A software engineering team at a mid-sized tech company analysed **500 software projects** over one year. They want to discover which **software quality metrics and defect patterns** commonly occur together — helping them predict and prevent software failures early.

The following data was extracted from their **project quality logs**:

Item Set	Number of Projects
{High Complexity}	250
{Low Test Coverage}	220
{Code Duplication}	200
{Long Methods}	180
{High Coupling}	160
{High Complexity, Low Test Coverage}	150
{High Complexity, Code Duplication}	130
{Low Test Coverage, Code Duplication}	110
{Long Methods, High Complexity}	120
{High Coupling, Low Test Coverage}	100
{High Complexity, Low Test Coverage, Code Duplication}	80
{Long Methods, High Complexity, Low Test Coverage}	70
{High Coupling, High Complexity, Code Duplication}	60
{Long Methods, High Coupling, Low Test Coverage}	50

Context: What These Metrics Mean

- **High Complexity** — code is difficult to understand and maintain
- **Low Test Coverage** — insufficient unit/integration tests
- **Code Duplication** — same code repeated in multiple places
- **Long Methods** — functions doing too many things at once
- **High Coupling** — modules too dependent on each other

Core Metric Formulae

Association Rule Metrics — Use these in all calculations

$$\text{Support}(X \Rightarrow Y) = \frac{N_{X \cup Y}}{N}$$

$$\text{Confidence}(X \Rightarrow Y) = \frac{N_{X \cup Y}}{N_X}$$

$$\text{Lift}(X \Rightarrow Y) = \frac{N_{X \cup Y} \times N}{N_X \times N_Y}$$

where $N = 500$ (total projects), N_X = projects containing X, N_Y = projects containing Y

Lift $> 1 \Rightarrow$ positive correlation, Lift $\approx 1 \Rightarrow$ independent, Lift $< 1 \Rightarrow$ negative correlation

Part A — Support Calculation

20 Marks

Member 1 — Individual Handwritten Submission

Using the data table above, compute the **support** for each of the following four rules. Show all working steps clearly.

1. {High Complexity} ⇒ {Low Test Coverage}
2. {Low Test Coverage} ⇒ {Code Duplication}
3. {High Complexity, Low Test Coverage} ⇒ {Code Duplication}
4. {Long Methods, High Complexity} ⇒ {Low Test Coverage}

After computing, answer the following:

- If the quality team sets **minsup = 0.20**, which of the rules survive the threshold? List them clearly.
- Which rule has the **highest support**? What does this mean practically for the software development team?
- Why does it make **real-world sense** from a software engineering perspective that this particular metric combination appears most frequently together?

Part B — Confidence Calculation

20 Marks

Member 2 — Individual Handwritten Submission

Using the same dataset, compute the **confidence** for each of the following four rules. Show all working steps clearly.

1. {High Complexity} ⇒ {Low Test Coverage}
2. {Code Duplication} ⇒ {High Complexity}
3. {High Complexity, Low Test Coverage} ⇒ {Code Duplication}
4. {High Coupling} ⇒ {Low Test Coverage}

After computing, answer the following:

- If the quality team sets **minconf = 0.55**, which rules survive?
- Rules 1 and 4 both have **Low Test Coverage** as their conclusion. Which antecedent is a stronger predictor of Low Test Coverage and why?
- From a **software quality perspective**, does the confidence of {High Coupling} ⇒ {Low Test Coverage} make practical sense? Explain using real-world software engineering reasoning.

Part C — Lift Calculation

25 Marks

Member 3 — Individual Handwritten Submission

Using the same dataset, compute the **lift** for each of the following four rules. Show all working steps clearly.

1. {High Complexity} ⇒ {Low Test Coverage}
2. {Code Duplication} ⇒ {High Complexity}
3. {High Complexity, Low Test Coverage} ⇒ {Code Duplication}
4. {High Coupling} ⇒ {Low Test Coverage}

After computing, answer the following:

- For each rule, state whether the metrics are **positively correlated, negatively correlated, or independent**.
- Which rule shows the **strongest genuine association** between software quality metrics?
- One of these rules likely has **high confidence but lift close to 1** — identify it and explain what this means for the quality team's decision making.
- From a **software quality management perspective**, which metric combination should the team **prioritise fixing first** based on lift values? Justify your answer.

Part D — Applied Decision Making

35 Marks

Member 4 — Individual Handwritten Submission (Depends on Parts A, B, C)

The **Software Quality Manager** wants to implement one of these two intervention strategies:

Strategy A **Refactoring Sprint:** Prioritise fixing High Complexity + Low Test Coverage first — assigning dedicated developers to reduce complexity and write more tests before any new features are built.

Strategy B **Code Review Policy:** Enforce strict code review rules targeting Code Duplication + High Coupling — any pull request showing these metrics together gets automatically rejected until fixed.

Task 1 — Complete Summary Table
Using values calculated by Members 1, 2 and 3, fill in the table below:

Rule	Support	Confidence	Lift	Passes Both?
{High Complexity} ⇒ {Low Test Coverage}				
{Code Duplication} ⇒ {High Complexity}				
{High Complexity, Low Test Coverage} ⇒ {Code Duplication}				
{High Coupling} ⇒ {Low Test Coverage}				
{Long Methods, High Complexity} ⇒ {Low Test Coverage}				

Thresholds: minsup = 0.20 and minconf = 0.55. Mark each row Yes or No in the last column.

Task 2 — Strategic Decision
Based on your completed table, which strategy — **A or B** — does the data support more strongly? Justify your answer using **all three metrics** (support, confidence AND lift) — not just one.

Task 3 — Team Size Calculation
The company has **500 active developers** working across all projects next quarter. Based on the strongest rule your group identified:

- How many projects would **likely exhibit** that metric combination? Show your calculation.
- If each project has an average of **3 developers**, how many developers should be **trained or reassigned** to address this quality issue? Show your calculation.

Task 4 — Manager Communication
The Quality Manager has **no technical background** in data mining. Write **3–4 sentences** explaining:

- What the strongest rule your group found actually **means in plain English**
- What the team should **do about it practically**
- What association rules **cannot tell them** — mention at least one limitation

Task 5 — New Rule Proposal
Based on your software engineering knowledge, propose **one new rule** not listed anywhere in this question that you believe would be worth testing. Specify:

- The **antecedent and consequent** of your proposed rule
- What **minsup and minconf** thresholds you would set and why
- Why you believe this rule would have a **high lift** value
- What **business action** the company would take if the rule is confirmed

TOTAL: 100 Marks

Important Notes for All Members

- All four parts in Question 02 are **connected** — Part D depends on Parts A, B, and C
- Show **all working steps** for every calculation
- Round all answers to **3 decimal places**
- Each member is individually responsible for their part but must ensure **consistency** with the group
- **Member 4 must not begin Task 1** until all values from Members 1, 2, and 3 are verified and agreed upon
- **F**or Groups having lesser or more members, they are required to adjust and manage the work in between all the members by themselves(at least one handwrritten part should be attempted by each member, for other parts they can write it on word file!) i.e., if there are three members, they will attempt three individual parts each and then for the fourth one, they have option to do it on word!

[Correction and Understanding]
`You will get 50% marks if everything is correct(In terms of answers)`
`and 50% marks will be of your demo(concepts and understanding)`
`On spot Solving of some Scenario can also be asked by TA to any member!`

Marks Summary

Part	Component	Marks
1	Dataset Selection	5
2	Distance Matrix	3
	Centroid Initialisation	2
	Optimal Number of Clusters	5
	Cluster Visualisation	5
	CSV Output	3
	Per-Cluster Interpretation	2
3	Agglomerative Clustering	8
	Divisive Clustering	8
	Agglomerative vs. Divisive Comparison	9
	Part A — Support (Member 1)	10
	Part B — Confidence (Member 2)	10
	Part C — Lift (Member 3)	10
	Part D — Applied Decision (Member 4)	10
	Grand Total	**90**

Reminder: Late submissions will NOT be accepted under any circumstances.

Zip naming: `member1-id_member2-id[_member3-id][_member4-id].zip`